

- **Epiphytes** grow on the surfaces of other plants but acquire water and minerals from rain. Parasitic plants absorb nutrients from host plants. Carnivorous plants supplement their mineral nutrition by digesting animals.

? Do all plants gain energy directly from photosynthesis? Explain.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- The inorganic nutrient most often lacking in crops is
(A) carbon. (C) phosphorus.
(B) nitrogen. (D) potassium.
- Micronutrients are needed in very small amounts because
(A) most of them are mobile in the plant.
(B) most serve mainly as cofactors of enzymes.
(C) most are supplied in large enough quantities in seeds.
(D) they play only a minor role in the growth and health of the plant.
- Mycorrhizae enhance plant nutrition mainly by
(A) absorbing water and minerals through the fungal hyphae.
(B) providing sugar to root cells, which have no chloroplasts.
(C) converting atmospheric nitrogen to ammonia.
(D) enabling the roots to parasitize neighboring plants.
- Epiphytes are
(A) fungi that attack plants.
(B) fungi that form mutualistic associations with roots.
(C) nonphotosynthetic parasitic plants.
(D) plants that grow on other plants.
- A problem with intensive irrigation is
(A) overfertilization.
(B) aquifer depletion.
(C) the long-term depletion of soil oxygen.
(D) the clogging of waterways by vegetation debris.

Levels 3-4: Applying/Analyzing

- A mineral deficiency is likely to affect older leaves more than younger leaves if
(A) the mineral is a micronutrient.
(B) the mineral is very mobile within the plant.
(C) the mineral is required for chlorophyll synthesis.
(D) the mineral is a macronutrient.
- The greatest difference in health between two groups of plants of the same species, one group with mycorrhizae and one group without mycorrhizae, would be in an environment
(A) where nitrogen-fixing bacteria are abundant.
(B) that has soil with poor drainage.
(C) that has hot summers and cold winters.
(D) in which the soil is relatively deficient in mineral nutrients.
- Two groups of tomatoes were grown under laboratory conditions, one with humus added to the soil and one a control without humus. The leaves of the plants grown without humus were yellowish (less green) compared with those of the plants grown in humus-enriched soil. The best explanation is that
(A) the healthy plants used the food in the decomposing leaves of the humus for energy to make chlorophyll.
(B) the humus made the soil more loosely packed, so water penetrated more easily to the roots.
(C) the humus contained minerals such as magnesium and iron needed for the synthesis of chlorophyll.
(D) the heat released by the decomposing leaves of the humus caused more rapid growth and chlorophyll synthesis.

- The specific relationship between a legume and its mutualistic *Rhizobium* strain probably depends on
(A) each legume having a chemical dialogue with a fungus.
(B) each *Rhizobium* strain having a form of nitrogenase that works only in the appropriate legume host.
(C) each legume being found where the soil has only the *Rhizobium* specific to that legume.
(D) specific recognition between chemical signals and signal receptors of the *Rhizobium* strain and legume species.

- DRAW IT** Draw a simple sketch of cation exchange, showing a root hair, a soil particle with anions, and a hydrogen ion displacing a mineral cation.

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Imagine taking the plant out of the picture in Figure 37.11. Write a paragraph explaining how soil bacteria could sustain the recycling of nitrogen *before* land plants evolved.
- SCIENTIFIC INQUIRY** Acid precipitation has an abnormally high concentration of hydrogen ions (H^+). One effect of acid precipitation is to deplete the soil of nutrients such as calcium (Ca^{2+}), potassium (K^+), and magnesium (Mg^{2+}). Suggest a hypothesis to explain how acid precipitation washes these nutrients from the soil. How might you test your hypothesis?
- SCIENCE, TECHNOLOGY, AND SOCIETY** In many countries, irrigation is depleting aquifers to such an extent that land is subsiding, harvests are decreasing, and it is becoming necessary to drill wells deeper. In many cases, the withdrawal of groundwater has now greatly surpassed the aquifers' rates of natural recharge. Discuss the possible consequences of this trend. What can society and science do to help alleviate this growing problem?
- WRITE ABOUT A THEME: INTERACTIONS** The soil in which plants grow teems with organisms from every taxonomic kingdom. In a short essay (100–150 words), discuss examples of how the mutualistic interactions of plants with bacteria, fungi, and animals improve plant nutrition.
- SYNTHESIZE YOUR KNOWLEDGE**



Making a footprint in the soil seems like an insignificant event. In a short essay (100–150 words), explain how a footprint would affect the properties of the soil and how these changes would affect soil organisms and the emergence of seedlings.

For selected answers, see Appendix A.

Explore Scientific Papers with Science in the Classroom 

How do parasitic plants communicate with their hosts?

Go to "Getting to Know Your Neighbors"
at www.scienceintheclassroom.org.

➔ **Instructors:** Questions can be assigned in Mastering Biology.